Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese

Department of Electrical, Electronic and Computer Engineering

Eksamen
VRAESTEL
Kopiereg voorbehou

Exam
QUESTION PAPER

Module EBN111
Elektrisiteit en Elektronika
Junie 2017

Module EBN111
Electricity and Electronics
June 2017

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

MEMO.

Eksamensinligting: ***Exam information:***			
Maksimum punte: *Maximum marks:*	92	Volpunte: *Full marks:*	90
Duur van vraestel: *Duration of paper:*	180 min + 20 min to complete OCR form *180 min + 20 min om OKH vorm te voltooi*	Oopboek / toeboek: *Open / closed book:*	Toe *Closed*
Totale aantal bladsye (hierdie blad ingesluit): *Total numer of pages (including this page):*		~~18~~ 34 + OCR	

BELANGRIK- ***IMPORTANT***

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
   The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die Optiese Karakter Herkenning (OKH) vorm ingevul word.
   All questions must be answered on the Optical Character Recognition (OCR) form.
3. Neem noukeurig kennis van die instruksies op die OKH vorm.
   Take careful consideration of the instructions on the OCR form

Interne Eksaminator(e): *Internal examiner(s):*	1. D le Roux 2. D Ramotsoela 3. X Ye
Eksterne eksaminator: External examinor:	J. Joubert

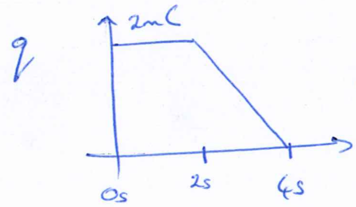
SECTION / AFDELING 1 [31]

Question 1 & 2 [3]	Vraag 1 & 2 [3]
01) An office building uses 100 lights of 80 W each. If they leave the lights on all day and night, what is the electricity account after one month (30 days)? Assume electricity costs R1.5 per kWh.	01) ‘n Kantoorgebou maak gebruik van 100 ligte van 80 W elk. Indien die ligte elke dag en nag aan is, wat is die koste van elektrisiteit na een maand (30 dae)? Aanvaar elektrisiteit kos R1.5 per kWh.
02) The charge through an element is constant at 2 mC between $t = 0$ s and $t = 2$ s. Between $t = 2$ s and $t = 4$ s, the charge falls linearly to 0 mC. What is the energy usage associated with the element if the potential difference across the element is 9 V?	02) Die lading deur ‘n element is ‘n konstante 2 mC tussen $t = 0$ s en $t = 2$ s. Tussen $t = 2$ s en $t = 4$ s, val die lading linêer na 0 mC. Wat is die energie verbuik geassosieer met die element indien die potensiaalverskil oor die element 9 V is?
01 R	**01** R
02 *W*	**02** *W*

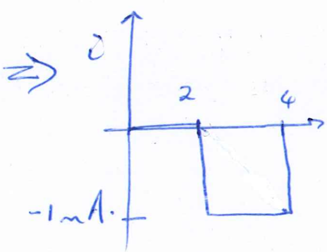
01 $\quad R = 100 \times 0.08\,\text{kW} \times (30 \times 24)\,\text{h} \times 1.5\ \text{R/kWh}$

$= \underline{R\,8640}$

0.2



$\Rightarrow \quad i = \dfrac{dq}{dt} \quad \Rightarrow$

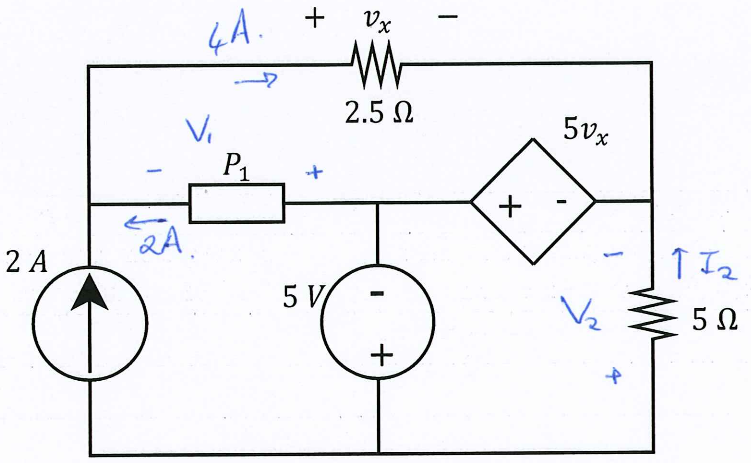


$$w = \int_0^4 v \cdot i \, dt$$

$$= 9\,(2 \times -1\,\text{mA})$$

$$= \underline{-18\ \text{mJ}}$$

Question 3 & 4 [4]		Vraag 3 & 4 [4]	
Determine the power associated with the dependent source and the unknown element P_1 if $v_x = 10$ V.		Bepaal die drywing geassosieer met die afhanklike bron en die onbekende element P_1 indien $v_x = 10$ V.	
03	$P_{(dep\text{-}source)}$	**03**	$P_{(afh\text{-}bron)}$
04	P_1	**04**	P_1



KVL: $+v_x - 5v_x + V_1 = 0$

$V_1 = +4v_x$

$= +40V.$

Power: $P_1 = +v \cdot i$

$= (40)(2) = \underline{+80\,W}.$

KVL: $+5 + 5v_x - V_2 = 0$

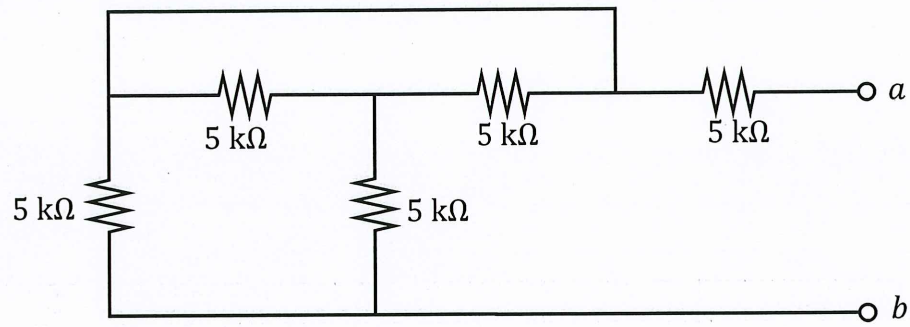
$V_2 = 55\,V.$

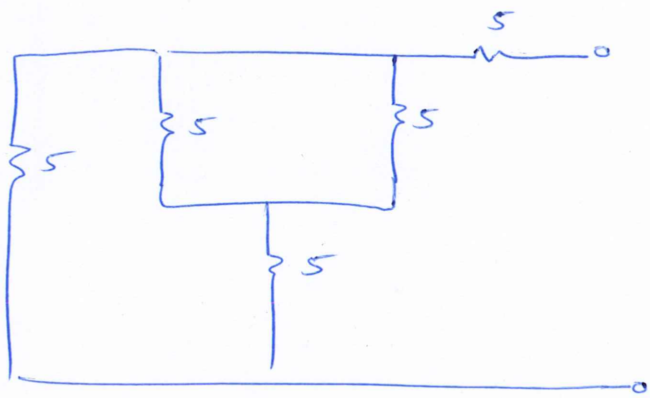
Ω: $I_2 = 55/5 = 11\,A.$

Power: $P_\Delta = -(5v_x)(15)$

$= \underline{-750\,W}.$

Question 5 [2]		Vraag 5 [2]	
05	Determine the equivalent resistance R_{ab} w.r.t. terminals *a*-*b*.	**05**	Bepaal ekwivalente weerstand R_{ab} met verwysing tot terminale *a* - *b*.





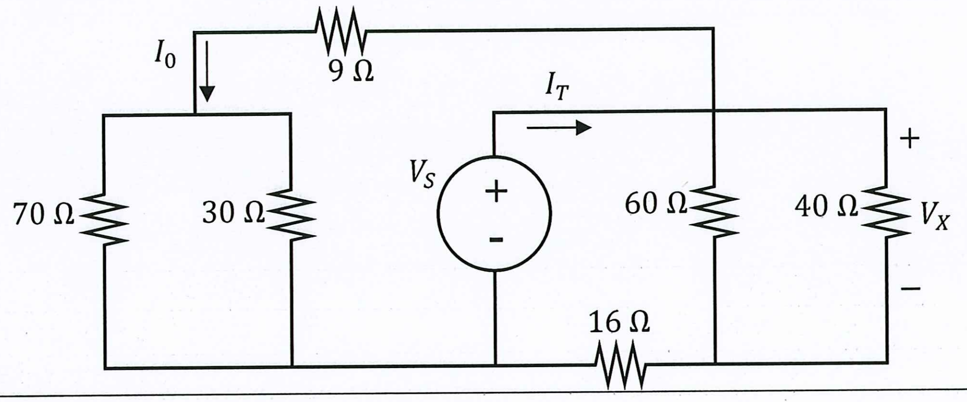
$$R_{ab} = 5 + 5 \| (5 + 2.5)$$

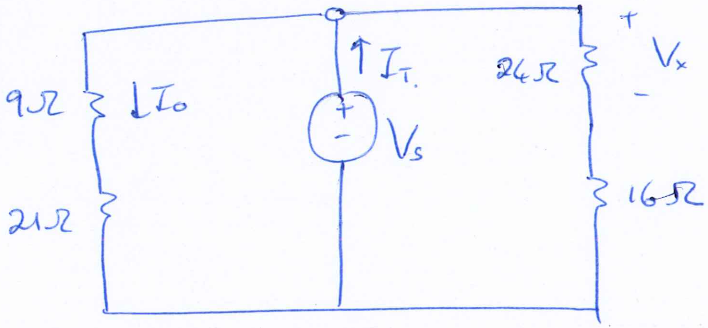
$$= 5 + 5 \| 7.5$$

$$= 5 + 3$$

$$= \underline{8 k\Omega}$$

Question 6 & 7 [4]		Vraag 6 & 7 [4]	
06	Determine V_S if $V_X = 9$ V.	**06**	Bepaal V_S indien $V_X = 9$ V.
07	Determine I_T if $I_0 = 2$ A.	**07**	Bepaal I_T indien $I_0 = 2$ A.





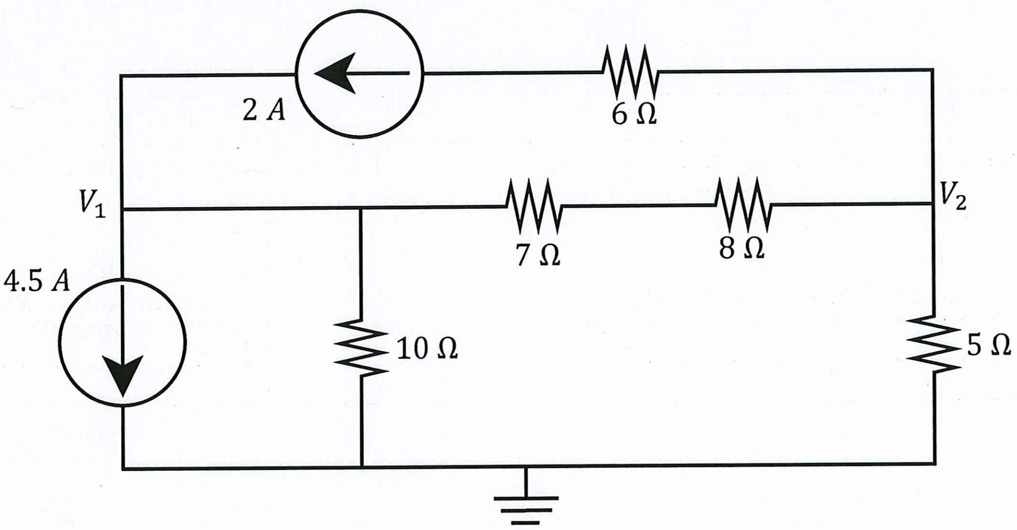
→ $24 = 60 \| 40$

→ $21 = 70 \| 30$

06 $\quad V_x = \dfrac{V_s \times 24}{24+16} \Rightarrow V_s = \dfrac{9\times 40}{24} = \underline{15\,V}$.

07 $\quad I_o = \dfrac{I_T \times 40}{40+30} \Rightarrow I_T = \dfrac{2\times 70}{40} = \underline{3.5\,A}$.

Question 8-11 [4]				Vraag 8-11 [4]			
Use nodal analysis to set up two equations of the following form: $$aV_1 + bV_2 = -75$$ $$cV_1 + dV_2 = \cancel{30}\ -30$$ Calculate the unknown coefficients.				Gebruik nodeanalise om twee vergelykings in die volgende vorm op te stel: $$aV_1 + bV_2 = -75$$ $$cV_1 + dV_2 = \cancel{30}\ -30.$$ Bepaal die onbekende koëffisiënte.			
08	*a*	**09**	*b*	**08**	*a*	**09**	*b*
10	*c*	**11**	*d*	**10**	*c*	**11**	*d*



① $$2 - 4.5 = \frac{V_1}{10} + \frac{V_1 - V_2}{15}$$

(× 30) $$-75 = 3V_1 + 2V_1 - 2V_2.$$

$$-75 = \underline{5}V_1 - \underline{2}V_2$$

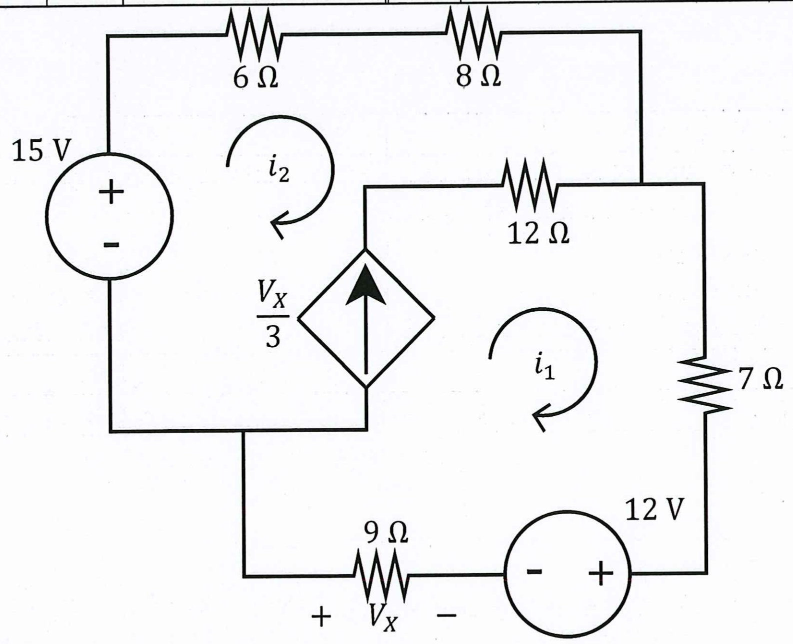
② $$-2 = \frac{V_2}{5} + \frac{V_2 - V_1}{15}$$

(× 15) $$-30 = 3V_2 + V_2 - V_1$$

$$-30 = \underline{4}V_2 - \underline{V_1}$$

Question 12-15 [4]	Vraag 12-15 [4]
Use mesh analysis to set up two equations of the following form: $$ei_1 + fi_2 = 3$$ $$gi_1 + hi_2 = 0$$ Calculate the unknown coefficients.	Gebruik lusanalise om twee vergelykings van die volgende vorm op te stel: $$ei_1 + fi_2 = 3$$ $$gi_1 + hi_2 = 0$$ Bepaal die onbekende koëffisiënte.

12	e	**13**	f	**12**	e	**13**	f
14	g	**15**	h	**14**	g	**15**	h



$V_x = -9i_1.$

① $-15 + 14i_2 + 16i_1 + 12 = 0.$

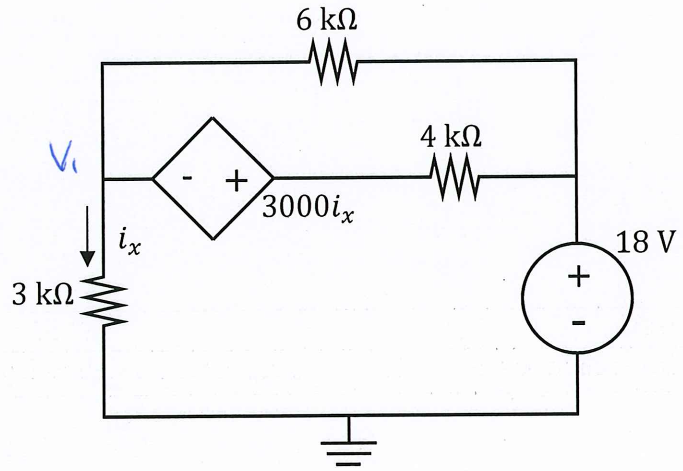
$\underline{\underline{16}}i_1 + \underline{\underline{14}}i_2 = 3$

② $\frac{V_x}{3} = i_1 - i_2.$

$-3i_1 = i_1 - i_2.$

$0 = \underline{\underline{4}}i_1 - \underline{\underline{i_2}}$

Question 16 [2]		Vraag 16 [2]	
16	Use either mesh or nodal analysis to solve for i_X.	16	Gebruik of node- of lus-analise om i_X te bereken.



$$\frac{V_1}{3k} + \frac{V_1 - 18}{6k} + \frac{V_1 + 3000 i_x - 18}{4k} = 0.$$

$(\times 12k)$ $\quad 4V_1 + 2V_1 - 36 + 3V_1 + 9000 i_x - 54 = 0$

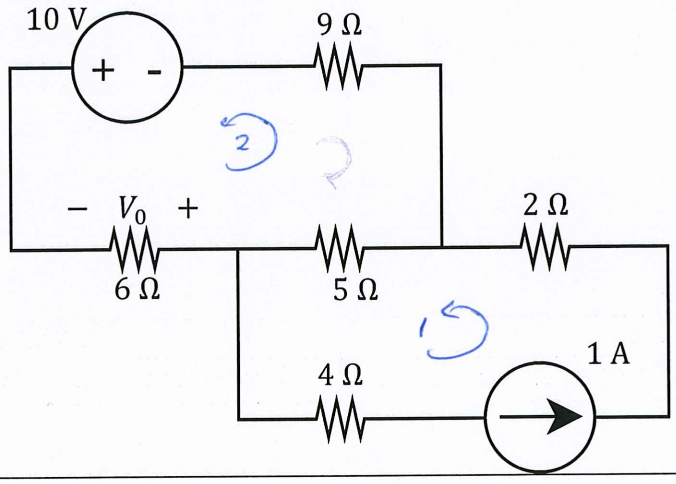
$$9V_1 + 9000 i_x = 90$$

but $\quad i_x = \frac{V_1}{3k} \rightarrow V_1 = 3000 i_x.$

$\therefore \quad 27000 i_x + 9000 i_x = 90$

$$i_x = \underline{2.5 mA}.$$

Question 17 [2]		Vraag 17 [2]	
17	Use either mesh or nodal analysis to solve for V_0.	**17**	Gebruik of node- of lus-analise om V_0 te bereken.



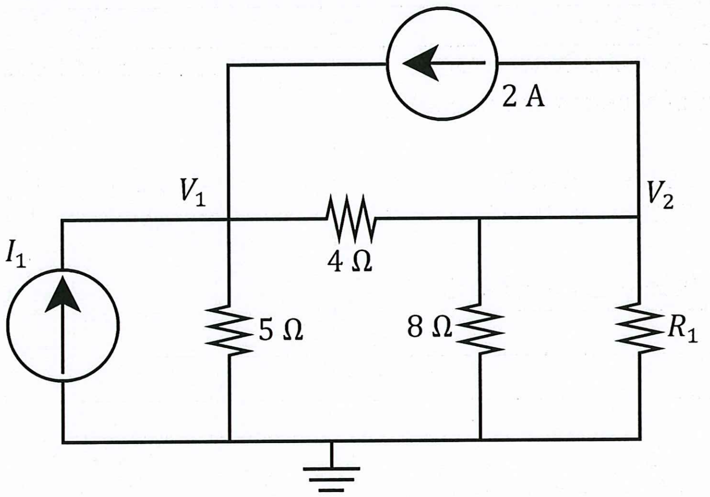
$i_1 = 1\text{ A}.$

$-10 + 20i_2 - 5i_1 = 0$

$i_2 = \frac{15}{20} = 0.75\text{ A}.$

$V_o = -6 \times i_2 = \underline{-4.5\text{ V}}.$

Question 18-19 [2]		Vraag 18-19 [2]	
The following matrix was obtained by means of inspection for the circuit below: $$\begin{bmatrix} 0.45 & -0.25 \\ -0.25 & 0.875 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 6 \\ -2 \end{bmatrix}$$ What is the value of I_I and R_I?		Die volgende matriks is vekry deur middel van inspeksie vir onderstaande kringbaan: $$\begin{bmatrix} 0.45 & -0.25 \\ -0.25 & 0.875 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 6 \\ -2 \end{bmatrix}$$ Wat is die waarde van I_I en R_I?	
18	I_I	**18**	I_I
19	R_I	**19**	R_I

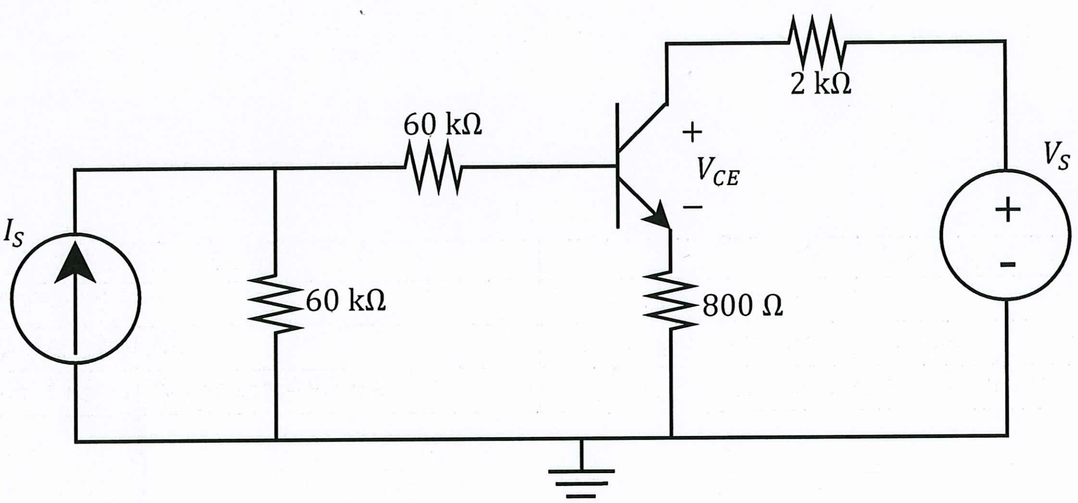


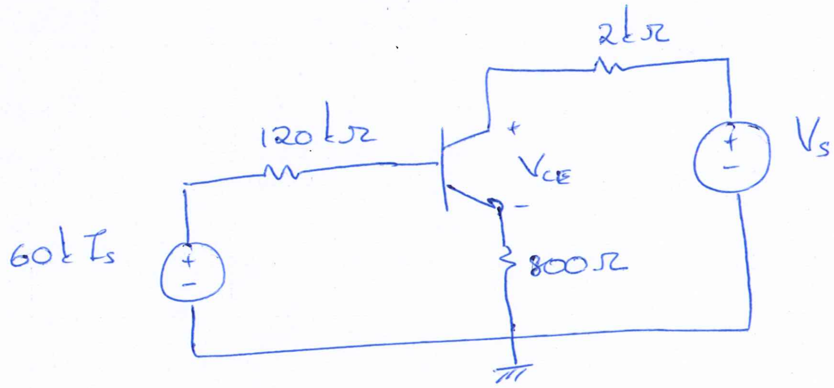
* $6 = 2 + I_1 \Rightarrow I_1 = \underline{4\,A}$

* $0.875 = \frac{1}{8} + \frac{1}{4} + \frac{1}{R_1}$

$R_1 = \underline{2\,\Omega}$

Question 20-21 [2]		Vraag 20-21 [2]	
$V_{BE} = 0.7\ V, \beta = 99.$		$V_{BE} = 0.7\ V, \beta = 99.$	
20	Find I_B if $I_S = 250\ \mu$A.	**20**	Bepaal I_B indien $I_S = 250\ \mu$A.
21	Find V_{CE} if $V_S = 20$ V and $I_C = 4.95\ m$A.	**21**	Bepaal V_{CE} indien $V_S = 20$ V en $I_C = 4.95\ m$A.





20

$$-60k(250\mu) + 120k\, I_B + 0.7 + 800(100)\, I_B = 0$$

$$I_B = \frac{14.3}{200k} = \underline{71.5\mu A}$$

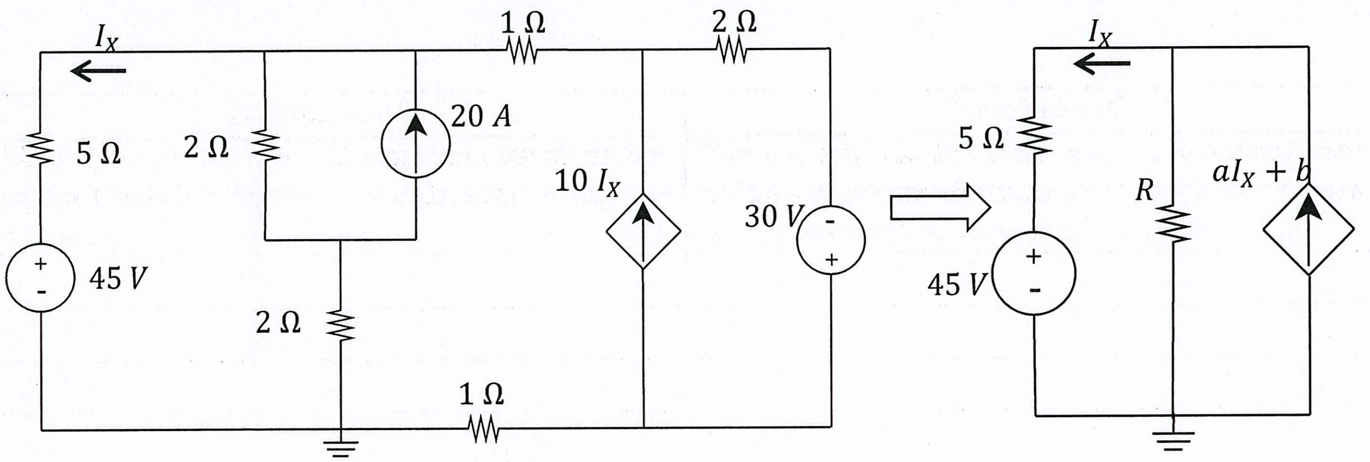
21

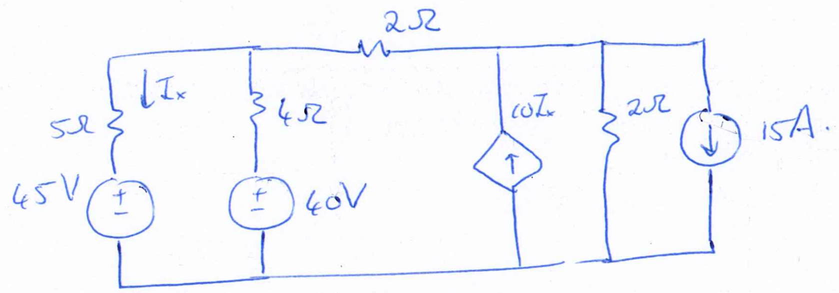
$$-20 + 2k\, I_C + V_{CE} + 800\left(\frac{\beta+1}{\beta}\right) I_C = 0.$$

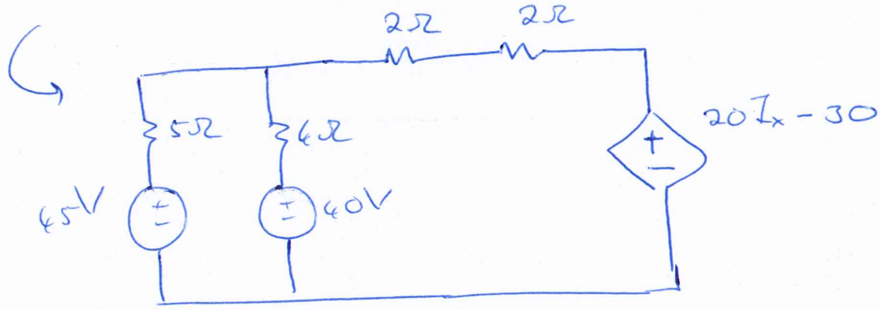
$$V_{CE} = 20 - 9.9 - 4 = \underline{6.1\,V}$$

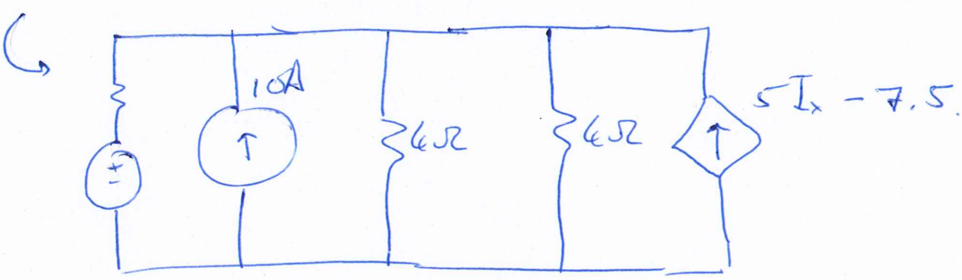
SECTION / AFDELING 2 [31]

Question 22-24 [4]		**Vraag 22-24 [4]**	
Simplify the circuit on the left with source transformation so that it looks like the one on the right, and determine the following:		Vereenvoudig die kringbaan aan die linkerkant deur middel van brontransformasie sodat dit lyk soos die een aan die regterkant, en bereken die volgende:	
22	R	**22**	R
23	a	**23**	a
24	b	**24**	b





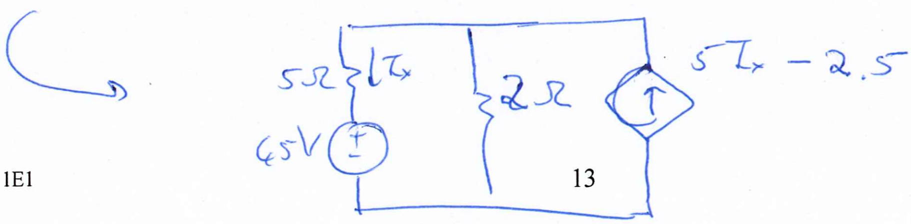




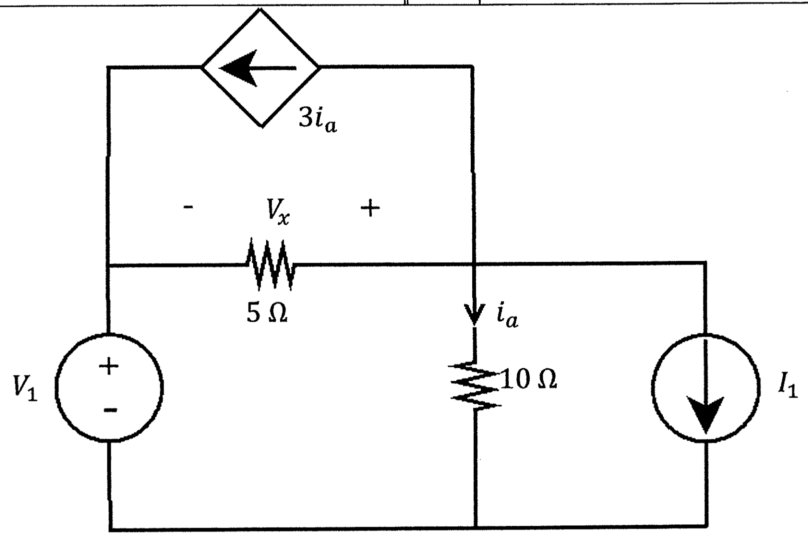
$R = 2$

$a = 5$

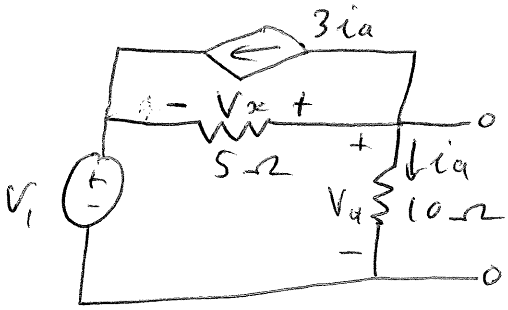
$b = -2.5$



Question 25 & 26 [4]		Vraag 25 & 26 [4]	
It is known that $V_x = 10$ V. The invidual contribution of the voltage source to V_x is 5 V. Determine:		Dit word gegee dat $V_x = 10$ V. Die individuele bydra van die spanningsbron tot V_x is 5 V. Bereken:	
25	V_1	**25**	V_1
26	I_1	**26**	I_1



$V_{V_x} = V_{ix} = 5V$



$3i_a + i_a = -\frac{V_x}{5}$

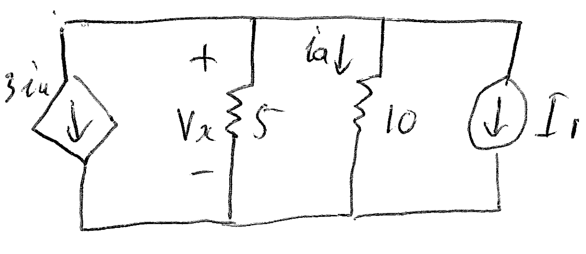
$4i_a = -1$

$i_a = -0{,}25$

$\therefore V_a = -2{,}5V$

$-V_1 - V_x + V_a = 0$

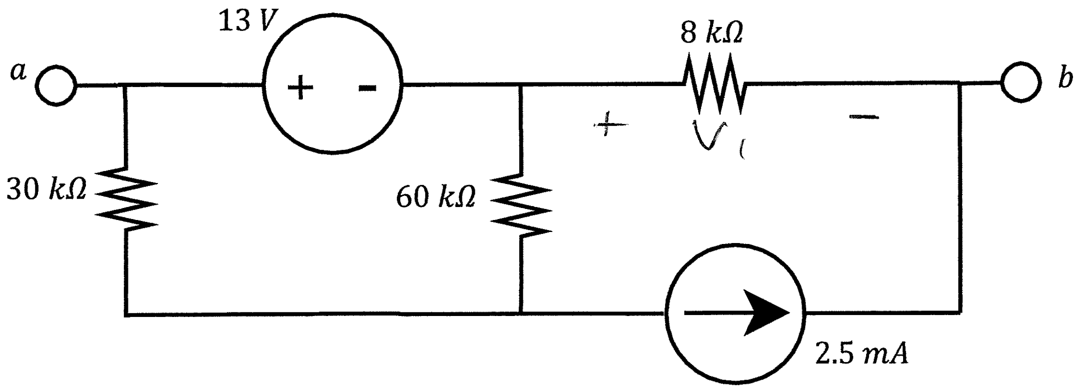
$\therefore \underline{V_1 = -7{,}5V}$



$3\left(\frac{5}{10}\right) + \frac{5}{5} + \frac{5}{10} + i_1 = 0$

$\therefore \underline{i_1 = -3A}$

Question 27 & 28 [4]		Vraag 27 & 28 [4]	
Determine the Thevenin equivalent voltage V_{TH} and resistance R_{TH} w.r.t. terminals *a-b*.		Bepaal die Thevenin ekwivalente spanning V_{TH} en weerstand R_{TH} m.v.t. terminale *a-b*	
27	V_{TH}	**27**	V_{TH}
28	R_{TH}	**28**	R_{TH}

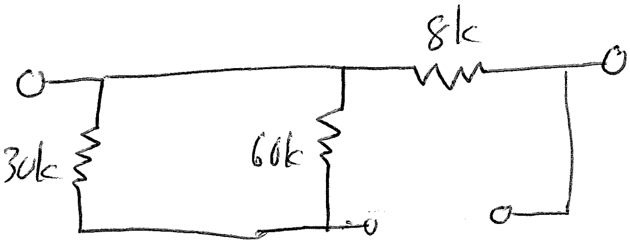


$$V_1 = -(2{,}5m)(8k) = -20V$$

$$V_{TH} = 13 + V_1$$

$$= 13 - 20$$

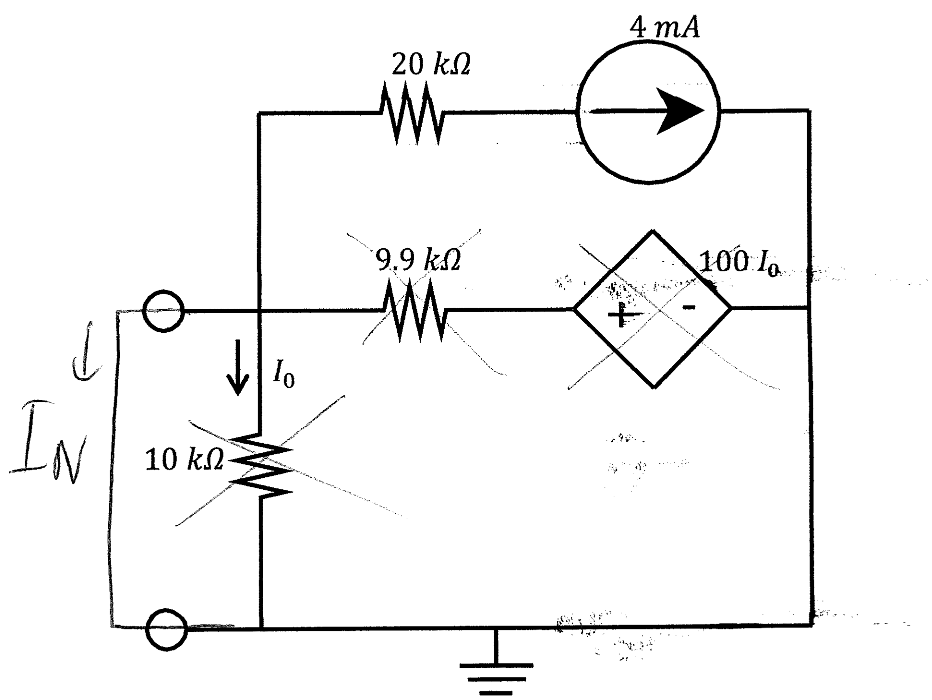
$$V_{TH} = -7V$$



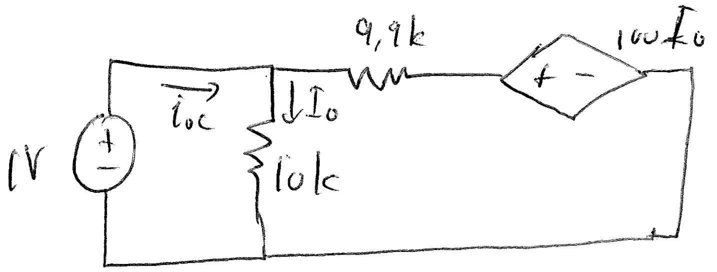
$$R_{TH} = 60k // 30k + 8k$$

$$R_{TH} = 28k\Omega$$

Question 29 & 30 [4]		Vraag 29 & 30 [4]	
Determine the Norton equivalent current I_N and resistance R_N w.r.t. terminals *a-b*.		Bepaal die Norton ekwivalente stroom I_N en weerstand R_N m.v.t. terminale *a-b*.	
29	I_N	29	I_N
30	R_N	30	R_N



$$I_N = -4mA$$



$$i_{oc} = \frac{1}{10k} + \frac{1 - \frac{100}{10k}}{9,9k}$$

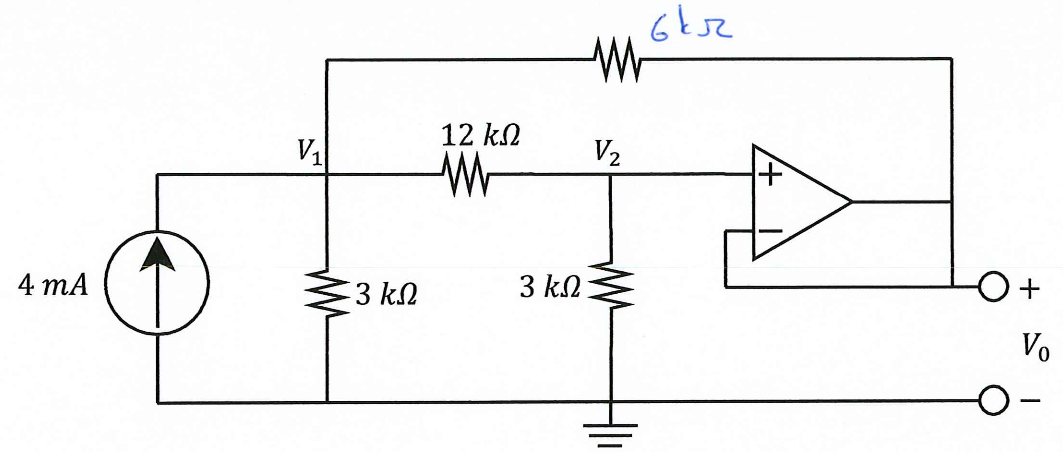
$$= \frac{1}{5000}$$

$$\therefore R_N = \frac{V_{oc}}{i_{oc}} = 5k\Omega$$

Question 31-34 [4]	Vraag 31-34 [4]
Use nodal analysis to set up two equations of the following form: $$aV_1 + bV_2 = 48$$ $$cV_1 + dV_2 = 0$$	Gebruik node-analise en skryf twee vergelykings in die volgende vorm:: $$aV_1 + bV_2 = 48$$ $$cV_1 + dV_2 = 0$$

31	a	32	b
33	c	34	d

31	a	32	b
33	c	34	d



① $\dfrac{V_1 - V_2}{6k} + \dfrac{V_1 - V_2}{12k} + \dfrac{V_1}{3k} = 4m$

$\times 12k$: $2V_1 - 2V_2 + V_1 - V_2 + 4V_1 = 48$

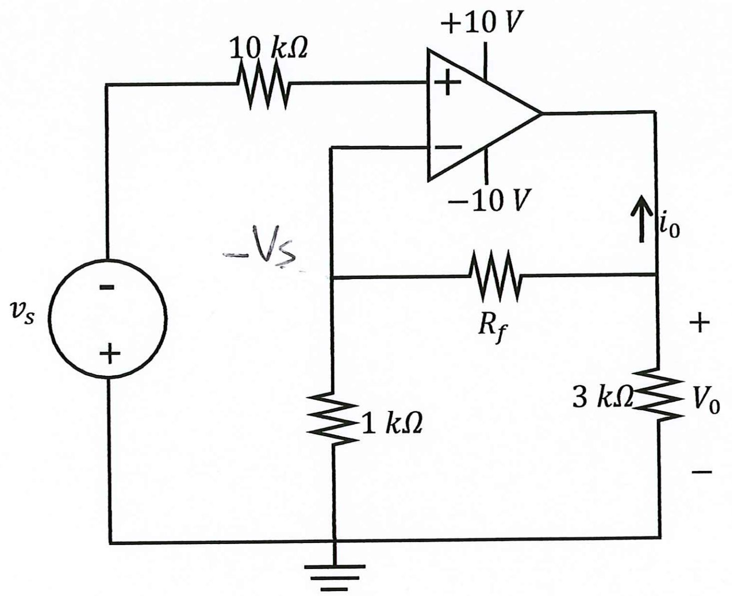
$7V_1 - 3V_2 = 48$; $a = 7$, $b = -3$

② $\dfrac{V_2 - V_1}{12k} + \dfrac{V_2}{3k} = 0$

$\times 12k$: $V_2 - V_1 + 4V_2 = 0$

$-V_1 + 5V_2 = 0$; $c = -1$; $d = 5$

Question 35 & 36 [4]		Vraag 35 & 36 [4]	
35	If $v_s = 2$ V, what value of the feedback resistance R_f will cause the operational amplifier to enter its negative saturation region?	**35**	Indien $v_s = 2$ V, vir watter waarde van die terugvoer weerstand R_f sal die operasionele versterker sy negatiewe versadigspunt bereik?
36	When $v_s = 6$ V and $R_f = 2$ kΩ, calculate the current i_o.	**36**	Indien $v_s = 6$ V end $R_f = 2$ kΩ, bereken die stroom i_o.



35) $$-\frac{V_s}{1k} + \frac{-V_s - V_o}{R_f} = 0$$

$$-\frac{2}{1k} + \frac{-2+10}{R_f} = 0$$

$$\therefore R_f = 4k\Omega$$

$$V_o = V_s\left(\frac{R_f}{1k}+1\right)$$

$$-10 = -2\left(\frac{R_f}{1k}+1\right)$$

$$R_f = 4k\Omega$$

36) $$-\frac{6}{1k} + \frac{-6-V_o}{2k} = 0$$

$$\times 2k: \quad -12 - 6 - V_o = 0$$

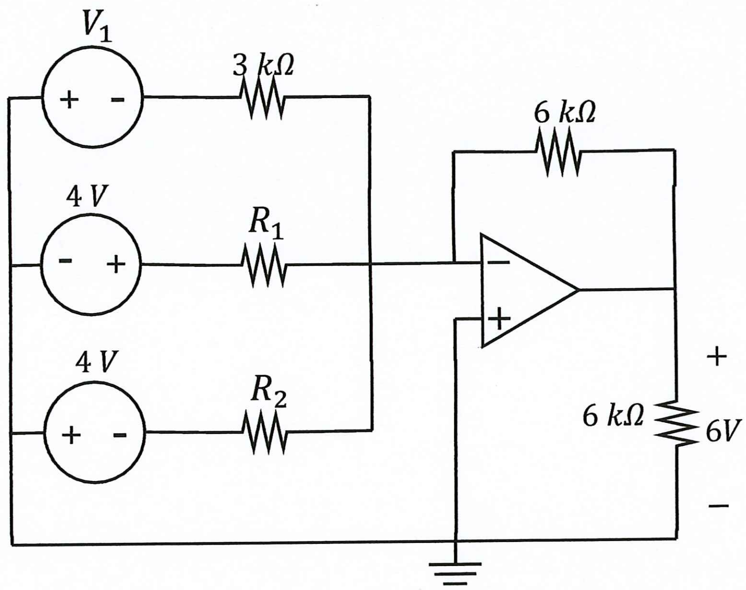
$$\therefore V_o = -18\text{ V}$$

$$i_o + \frac{V_o+6}{2k} + \frac{V_o}{3k} = 0$$

$$i_o - 0,006 - 0,006 = 0$$

$$\therefore i_o = 0,012\text{ A}$$

Question 37 & 38 [3]		Vraag 37 & 38 [3]	
37	Determine V_1 if $R_1 \to \infty$ and $R_2 \to \infty$.	37	Bepaal V_1 indien if $R_1 \to \infty$ en $R_2 \to \infty$.
38	Determine R_1 if $V_1 = 2$ V and $R_2 = 2$ kΩ.	38	Bepaal R_1 indien $V_1 = 2$ V en $R_2 = 2$ kΩ.



37) $6 = \left(-\frac{6k}{3k}\right)(-V_1)$

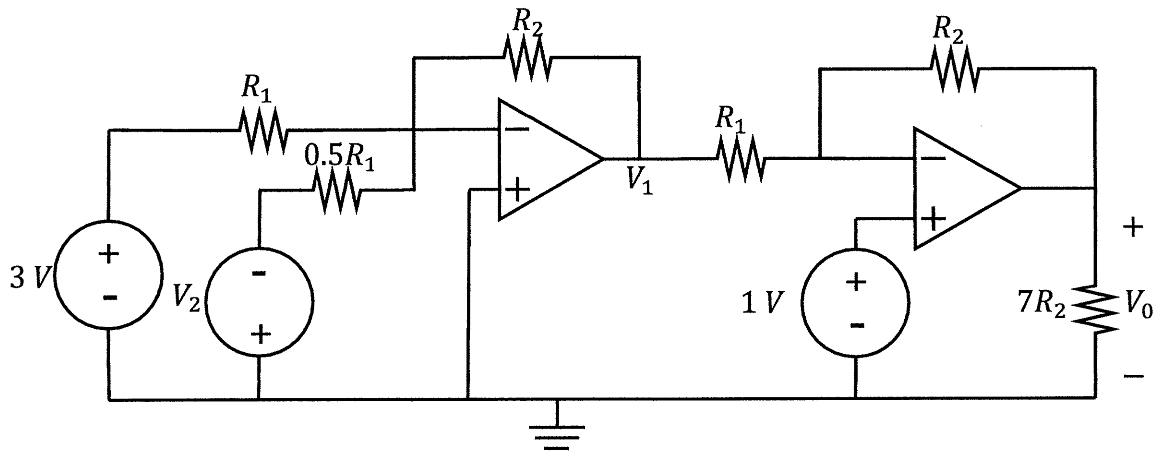
$\therefore V_1 = 3V$

38) $6 = -\left(\frac{6k}{3k}(-2) + \frac{6k}{R_1}(4) + \frac{6k}{2k}(-4)\right)$

$+10 = \frac{24k}{R_1}$

$\therefore R_1 = 2{,}4k\Omega$

Question 39 & 40 [4]		Vraag 39 & 40 [4]	
If $V_1 = 2$ V and $\frac{R_2}{R_1} = 2$, determine:		Indien $V_1 = 2$ V en $\frac{R_2}{R_1} = 2$, bepaal:	
39	V_2	**39**	V_2
40	V_0	**40**	V_0



39) $V_1 = -\left(\frac{R_2}{R_1}(3) + \frac{R_2}{0{,}5R_1}(-V_2)\right)$

$2 = -[(2)(3) + (2)(2)(-V_2)]$

$2 = -6 + 4V_2$

$\therefore \underline{V_2 = 2V}$

40) $\frac{1-V_1}{R_1} + \frac{1-V_0}{R_2} = 0$

$\times R_2$: $\frac{R_2}{R_1}(1-2) + 1 - V_0 = 0$

$(2)(-1) + 1 - V_0 = 0$

$\therefore \underline{V_0 = -1V}$

SECTION / AFDELING 3 [31]

Question 41 [2]		Vraag 41 [2]	
The voltage $v(t) = 6t^2 - 2t$ V is applied across a 100 mH inductor. Determine $i_L(t)$ at $t = 2s$ if $i_L(0) = 5$ A.		Die spanning $v(t) = 6t^2 - 2t$ V word oor 'n 100 mH induktor toegepas. Bepaal $i_L(t)$ by $t = 2s$ indien $i_L(0) = 5$ A.	
41	$i_L(2)$	41	$i_L(2)$

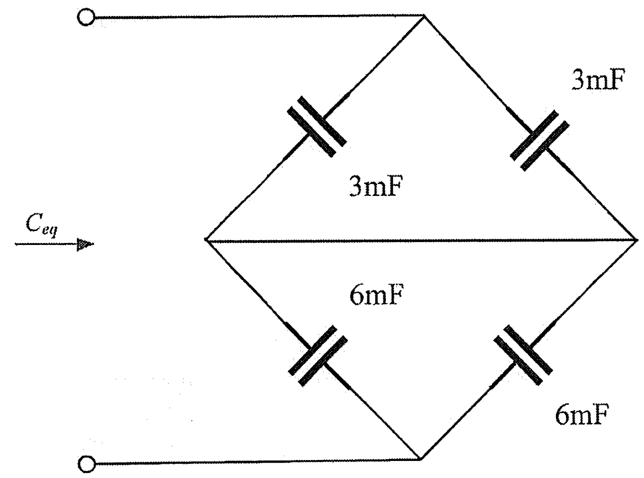
$$i_L(t) = \frac{1}{L}\int_0^t v(t)\,dt + i_L(0)$$

$$= 10\int_0^t (6t^2 - 2t)\,dt + 5$$

$$= 10(2t^3 - t^2) + 5$$

$$i_L(2) = 10(16-4) + 5 = \underline{125A}$$

Question 42 [2]		Vraag 42 [2]	
42	Determine C_{eq}.	42	Bepaal C_{eq}.

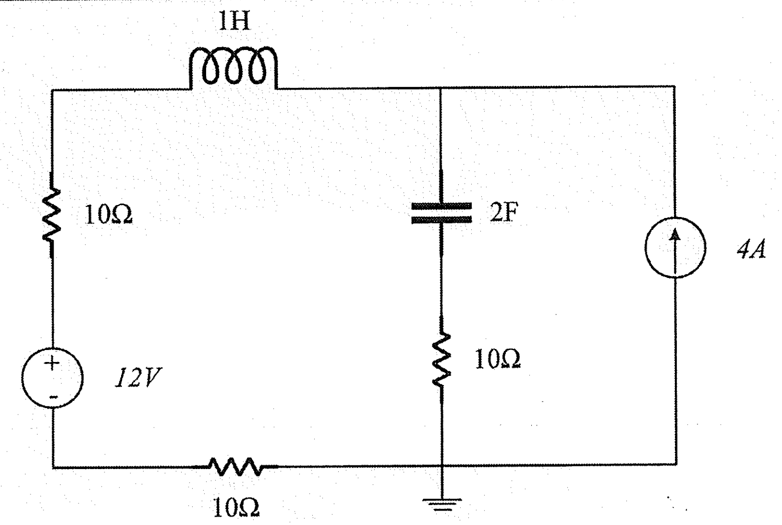


$$C_{eq} = (3+3) \,\&\, (6+6)$$

$$= \frac{6 \times 12}{6+12}$$

$$= \underline{4\ mF}$$

Question 43 [2]		Vraag 43 [2]	
43	Assume steady state conditions (DC conditions), and determine the energy E_C stored in the capacitor.	**43**	Aanvaar gestadigde toestand (GS toestande), en bepaal die energie E_C gestoor in die kapasitor.



$V_C = 12 + 4\times(10+10) = 92V$

$E_C = \frac{1}{2}CV_C^2 = \frac{1}{2}\times 2 \times 92^2 J = 8464 J$

Question 44-47 [4]		Vraag 44-47 [2]	
a) In Figure (a), assume: $v_1(t) = 4\cos(2\pi t + 30°)\ V$, $v_2(t) = 4\sin(2\pi t)$. Determine $\bar{Z}_2$ if $\bar{Z}_1 = 8\angle 120°\Omega$. Write the answer in polar form $A\angle\phi\ \Omega$, where $A > 0$ and $-180° \leq \phi \leq 180°$. b) In Figure (b), find the input impedance $\bar{Z}_T = Z_x + jZ_y$ at $\omega = 10^4 rad/s$.		a) Vir Figuur (a), aanvaar: $v_1(t) = 4\cos(2\pi t + 30°)\ V$, $v_2(t) = 4\sin(2\pi t)$. Bepaal $\bar{Z}_2$ indien $\bar{Z}_1 = 8\angle 120°\Omega$. Skryf die antwoord in polêre vorm $A\angle\phi\ \Omega$, met $A > 0$ en $-180° \leq \phi \leq 180°$. b) Vir Figuur (b), bepaal die insetimpedansie $\bar{Z}_T = Z_x + jZ_y$ by $\omega = 10^4 rad/s$.	
44	A [1]	**44**	A [1]
45	ϕ [1]	**45**	ϕ [1]
46	Z_x[1]	**46**	Z_x[1]
47	Z_y[1]	**47**	Z_y[1]

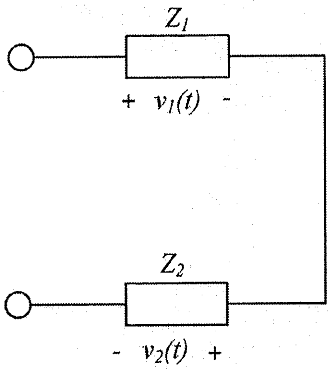


Figure (a)

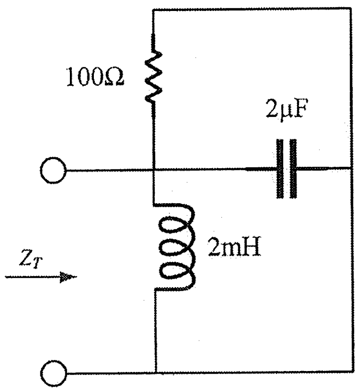


Figure (b)

$$\bar{V}_1 = 4\angle 30°\ V$$

$$\bar{V}_2 = 4\angle -90°\ V$$

$$\bar{I} = \frac{\bar{V}_1}{\bar{Z}_1} = \frac{4\angle 30°}{8\angle 120°}\ A$$

$$= 0.5\angle -90°\ A$$

$$\bar{Z}_2 = \frac{\bar{V}_2}{\bar{I}} = \frac{4\angle -90°}{0.5\angle -90°}\ \Omega$$

$$= 8\angle 0°\ \Omega$$

$$A = 8\ \Omega$$

$$\phi = 0°$$

$$Z_T = j\omega L \,||\, 100 \,||\, (-j\tfrac{1}{\omega C})$$

$$= j20 \,||\, 100 \,||\, (-j50)$$

$$= j20 \,||\, (20 - j40)$$

$$= 10 + j30\ \Omega$$

$$Z_x = 10\ \Omega$$

$$Z_y = 30\ \Omega.$$

Question 48-51 [4]		Vraag 48-51 [4]	
a) Simplify the following expression and write the answers in **rectangular form**. $a+jb=15\angle30^\circ\left(6-j8+\frac{2\angle60^\circ}{2+j4\sin(30^\circ)}\right)$ b) Determine $\bar{I}_2$ in **rectangular form** $(c+jd)$. $\begin{bmatrix}20\angle-60^\circ & -5\angle30^\circ\\16\angle0^\circ & 3\angle150^\circ\end{bmatrix}\begin{bmatrix}\bar{I}_1\\\bar{I}_2\end{bmatrix}=\begin{bmatrix}8+j6\\0\end{bmatrix}$		a) Vereenvoudig die volgende uitdrukking en skryf die antwoord in **reghoekige vorm**. $a+jb=15\angle30^\circ\left(6-j8+\frac{2\angle60^\circ}{2+j4\sin(30^\circ)}\right)$ b) Bepaal $\bar{I}_2$ in **reghoekige vorm** $(c+jd)$. $\begin{bmatrix}20\angle-60^\circ & -5\angle30^\circ\\16\angle0^\circ & 3\angle150^\circ\end{bmatrix}\begin{bmatrix}\bar{I}_1\\\bar{I}_2\end{bmatrix}=\begin{bmatrix}8+j6\\0\end{bmatrix}$	
48	a	**48**	a
49	b	**49**	b
50	c	**50**	c
51	d	**51**	d

$$a+jb=15\angle30^\circ\left(6-j8+\frac{2\angle60^\circ}{2+j2}\right)$$

$$=145.44-j51.423$$

$\underline{a=145.44}$

$\underline{b=-51.423}$

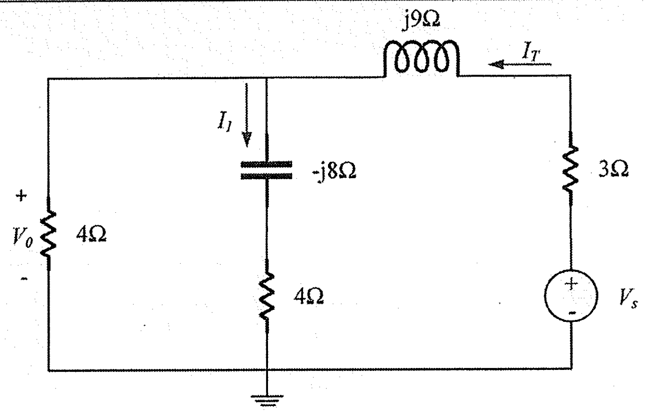
$$I_2=\frac{\begin{vmatrix}20\angle-60^\circ & 8+j6\\16\angle0^\circ & 0\end{vmatrix}}{\begin{vmatrix}20\angle-60^\circ & -5\angle30^\circ\\16\angle0^\circ & 3\angle150^\circ\end{vmatrix}}=\frac{-16(8+j6)}{60\angle90^\circ+80\angle30^\circ}$$

$$=\frac{-128-j96}{40\sqrt{3}+j100}=-1.2478+j0.41547$$

$\underline{c=-1.2478}$

$\underline{d=0.41547}$

Question 52-55 [4]		Vraag 52-55 [4]	
a) Find $\bar{I}_1 = I_x + jI_y$ A if $\bar{I}_T = 10\angle 0°$A. b) Find $\bar{V}_o = V_x + jV_y$ V if $\bar{V}_s = 100\angle 30°$V. (Give the answers in **rectangular form**.)		a) Bepaal $\bar{I}_1 = I_x + jI_y$ A indien $\bar{I}_T = 10\angle 0°$A. b) Bepaal $\bar{V}_o = V_x + jV_y$ V indien $\bar{V}_s = 100\angle 30°$V. (Give the answers in **rectangular form**.)	
52	I_x	**52**	I_x
53	I_y	**53**	I_y
54	V_x	**54**	V_x
55	V_y	**55**	V_y



$$I_1 = I_T \cdot \frac{4}{8-j8} = 10\angle 0° \cdot \frac{4}{8-j8} = 2.5 + j2.5\ A$$

$I_x = 2.5\ A$, $\quad I_y = 2.5\ A$

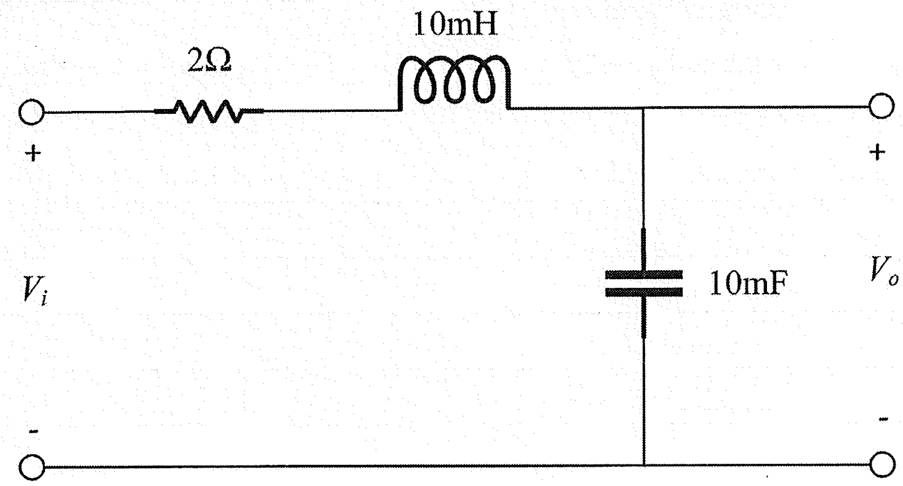
$$Z_T = 4 \,\|\, (4-j8) = \frac{4(4-j8)}{8-j8} = 3-j\ \Omega$$

$$\bar{V}_o = V_s \cdot \frac{Z_T}{Z_T+3+j9} = V_s \cdot \frac{3-j}{6+j8} = 23.660 - j20.981\ (V)$$

$V_x = 23.660\ V$

$V_y = -20.981\ V$

Question 56-59 [3]		Vraag 56-59 [3]	
Find the ratio $\frac{\bar{V}_o}{\bar{V}_i}$ for: 1. $\omega = 0$. (Give answer as $a = \frac{\bar{V}_o}{\bar{V}_i}\|_{\omega=0}$) 2. $\omega \to \infty$. (Give answer as $b = \frac{\bar{V}_o}{\bar{V}_i}\|_{\omega\to\infty}$) 3. $\omega = 100rad/s$. (Give answer: $c + jd = \frac{\bar{V}_o}{\bar{V}_i}\|_{\omega=100}$.)		Bepaal die verhouding $\frac{\bar{V}_o}{\bar{V}_i}$ vir: 1. $\omega = 0$. (Gee antwoord as $a = \frac{\bar{V}_o}{\bar{V}_i}\|_{\omega=0}$) 2. $\omega \to \infty$. (Gee antwoord as $b = \frac{\bar{V}_o}{\bar{V}_i}\|_{\omega\to\infty}$) 3. $\omega = 100rad/s$. (Gee antwoord: $c + jd = \frac{\bar{V}_o}{\bar{V}_i}\|_{\omega=100}$.)	
56	$a = \frac{\bar{V}_o}{\bar{V}_i}\|_{\omega=0}$	**56**	$a = \frac{\bar{V}_o}{\bar{V}_i}\|_{\omega=0}$
57	$b = \frac{\bar{V}_o}{\bar{V}_i}\|_{\omega\to\infty}$	**57**	$b = \frac{\bar{V}_o}{\bar{V}_i}\|_{\omega\to\infty}$
58	$c = R_e\left(\frac{\bar{V}_o}{\bar{V}_i}\|_{\omega=100}\right)$	**58**	$c = R_e\left(\frac{\bar{V}_o}{\bar{V}_i}\|_{\omega=100}\right)$
59	$d = I_m\left(\frac{\bar{V}_o}{\bar{V}_i}\|_{\omega=100}\right)$	**59**	$d = I_m\left(\frac{\bar{V}_o}{\bar{V}_i}\|_{\omega=100}\right)$



$$\frac{V_o}{V_i} = \frac{1/j\omega C}{R + j\omega L + 1/j\omega C} = \frac{1}{(1-\omega^2 LC) + j\omega RC}$$

$\omega = 0$, $\frac{V_o}{V_i} = \underline{a = 1}$

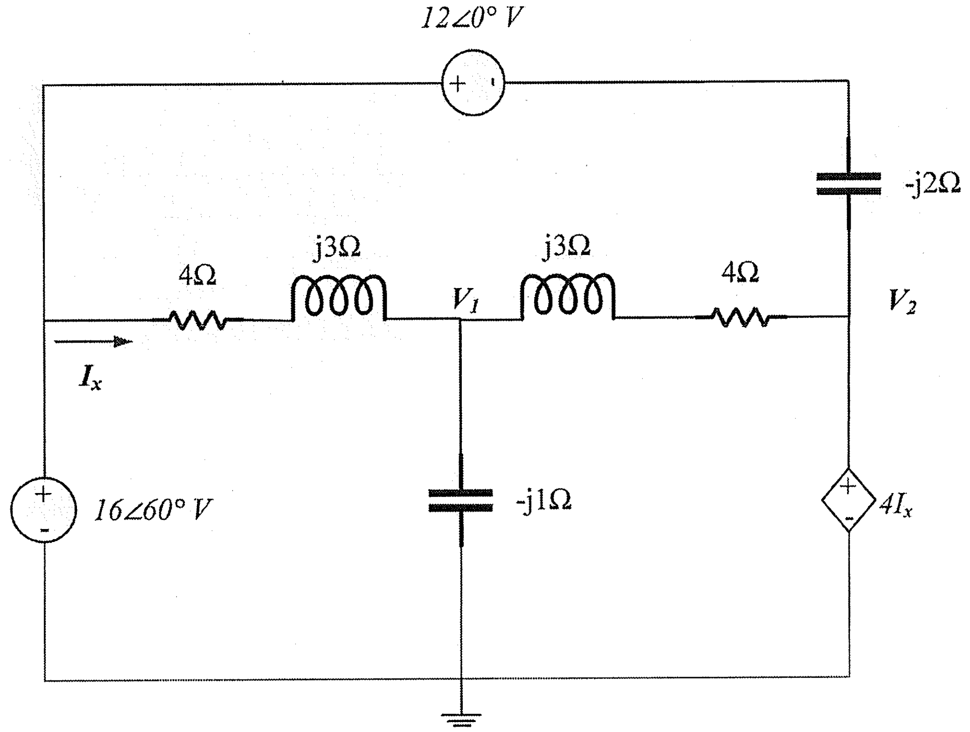
$\omega \to \infty$, $\frac{V_o}{V_i} = \underline{b = 0}$

$\omega = 100$, $\frac{V_o}{V_i} = \frac{1}{(1-10^4 \times 10 \times 10^{-3} \times 10 \times 10^{-3}) + j100 \times 2 \times 10 \times 10^{-3}} = -j0.5$

$\underline{c = 0}$

$\underline{d = -0.5}$

Question 60-65 [4]		Vraag 60-65 [4]	
Use nodal analysis to set up two equations of the following form: $$4\bar{V}_1 + (4+j3)\bar{V}_2 = A\angle\phi^\circ$$ $$a\bar{V}_1 + b\bar{V}_2 = 16\angle 60^\circ$$ where $A > 0$, $-180^\circ \le \phi \le 180^\circ$, $a = x_1 + jy_1$ and $b = x_2 + jy_2$ in **rectangular form**.		Gebruik node-analise om twee vergelykings in die volgende vorm op te stel: $$4\bar{V}_1 + (4+j3)\bar{V}_2 = A\angle\phi^\circ$$ $$a\bar{V}_1 + b\bar{V}_2 = 16\angle 60^\circ$$ met $A > 0$, $-180^\circ \le \phi \le 180^\circ$, $a = x_1 + jy_1$ en $b = x_2 + jy_2$ in **rectangular form**.	
60	A	**60**	A
61	ϕ	**61**	ϕ
62	x_1	**62**	x_1
63	y_1	**63**	y_1
64	x_2	**64**	x_2
65	y_2	**65**	y_2



① $V_2 = 4I_x = 4 \cdot \dfrac{16\angle 60^\circ - V_1}{4+j3} \Rightarrow (4+j3)V_2 = 64\angle 60^\circ - 4V_1$

$\Rightarrow 4V_1 + (4+j3)V_2 = 64\angle 60^\circ$, $A = 64$, $\phi = 60^\circ$

② $\dfrac{16\angle 60^\circ - V_1}{4+j3} = \dfrac{V_1 - V_2}{4+j3} + \dfrac{V_1}{-j1}$

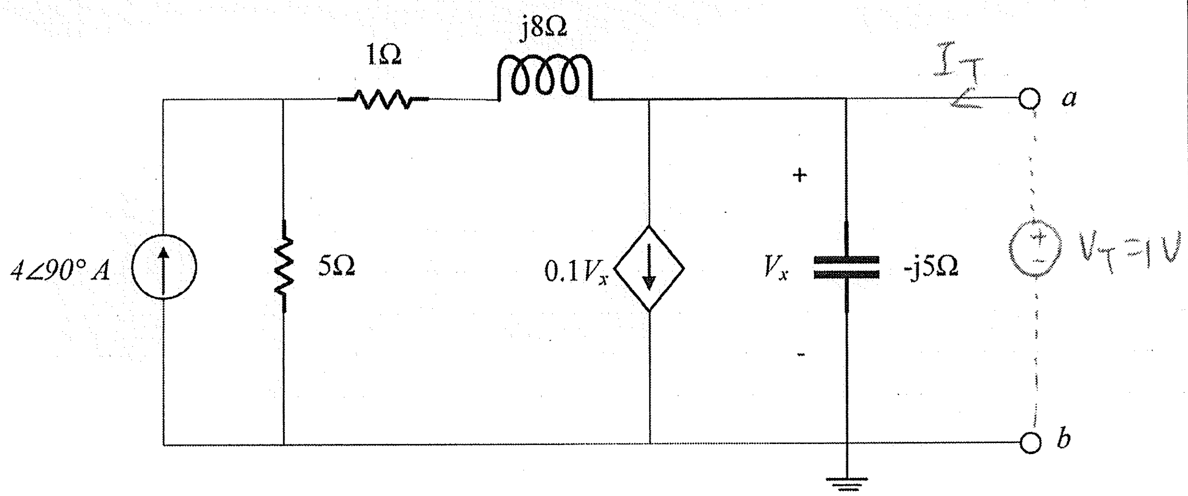
$x_1 = -1$, $y_1 = 4$

$16\angle 60^\circ - V_1 = V_1 - V_2 + j(4+j3)V_1$

$x_2 = -1$, $y_2 = 0$

$\Rightarrow (-1+j4)V_1 - V_2 = 16\angle 60^\circ$

Question 66-69 [4]		Vraag 66-69 [4]	
Find the Norton equivalent with respect to terminals *a*-*b*. Write the answers in **rectangular form**, where $\bar{I}_N = I_x + jI_y$A and $Z_N = Z_x + jZ_y\ \Omega$.		Bepaal die Norton ekwivalent met verwysing tot terminale *a*-*b*. Gee die antwoorde in **reghoekige formaat** met $\bar{I}_N = I_x + jI_y$ A en $Z_N = Z_x + jZ_y\ \Omega$.	
66	I_x	**66**	I_x
67	I_y	**67**	I_y
68	Z_x	**68**	Z_x
69	Z_y	**69**	Z_y



$$I_N = 4\angle 90^\circ \cdot \frac{5}{6+j8} = 1.6 + j1.2 \text{ A}$$

$I_x = 1.6$A, $\quad I_y = 1.2$A

Let $V_T = V_x = 1$V,

$$I_T = 0.1V_x + \frac{V_x}{-j5} + \frac{V_x}{(6+j8)}$$

$$= 0.1 + j0.2 + 0.06 - j0.08$$

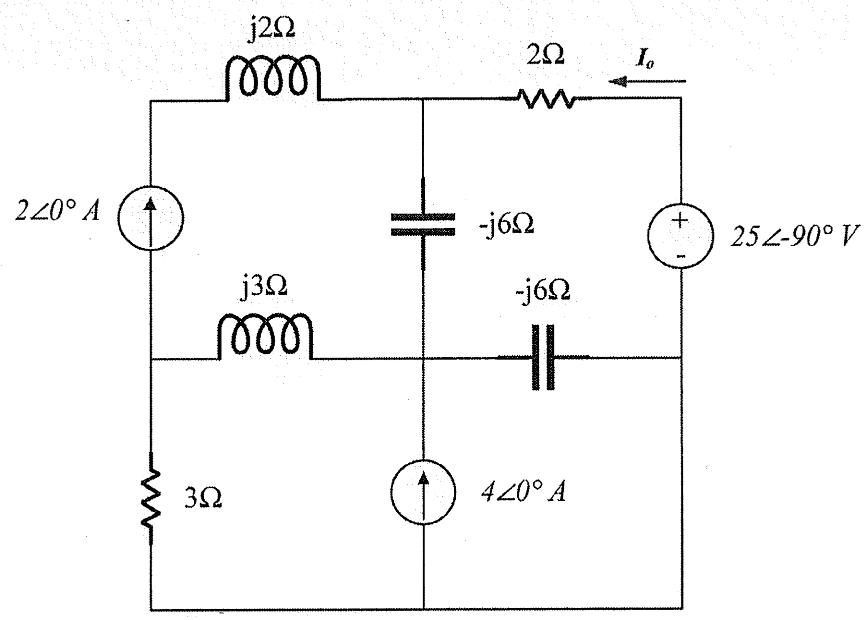
$$= 0.16 + j0.12$$

$$Z_N = \frac{V_T}{I_T} = \frac{1}{0.16+j0.12} = 4 - j3$$

$Z_x = 4\ \Omega$

$Z_y = -3\ \Omega$

Question 70-71 [2]		Vraag 70-71 [2]	
Determine $\bar{I}_o = I_x + jI_y$ A only due to the voltage source.		Bepaal $\bar{I}_o = I_x + jI_y$ A slegs as gevolg van die spanningsbron.	
70	I_x	**70**	I_x
71	I_y	**71**	I_y



$$I_o = \frac{25\angle -90^\circ}{Z_T},$$

$$Z_T = (2-j6) + (-j6)\,||\,(3+j3)$$

$$= 8 - j6\ \Omega.$$

$$I_o = \frac{25\angle -90^\circ}{8-j6} A = 1.5 - j2\ (A)$$

$I_x = 1.5$ A

$I_y = -2$ A